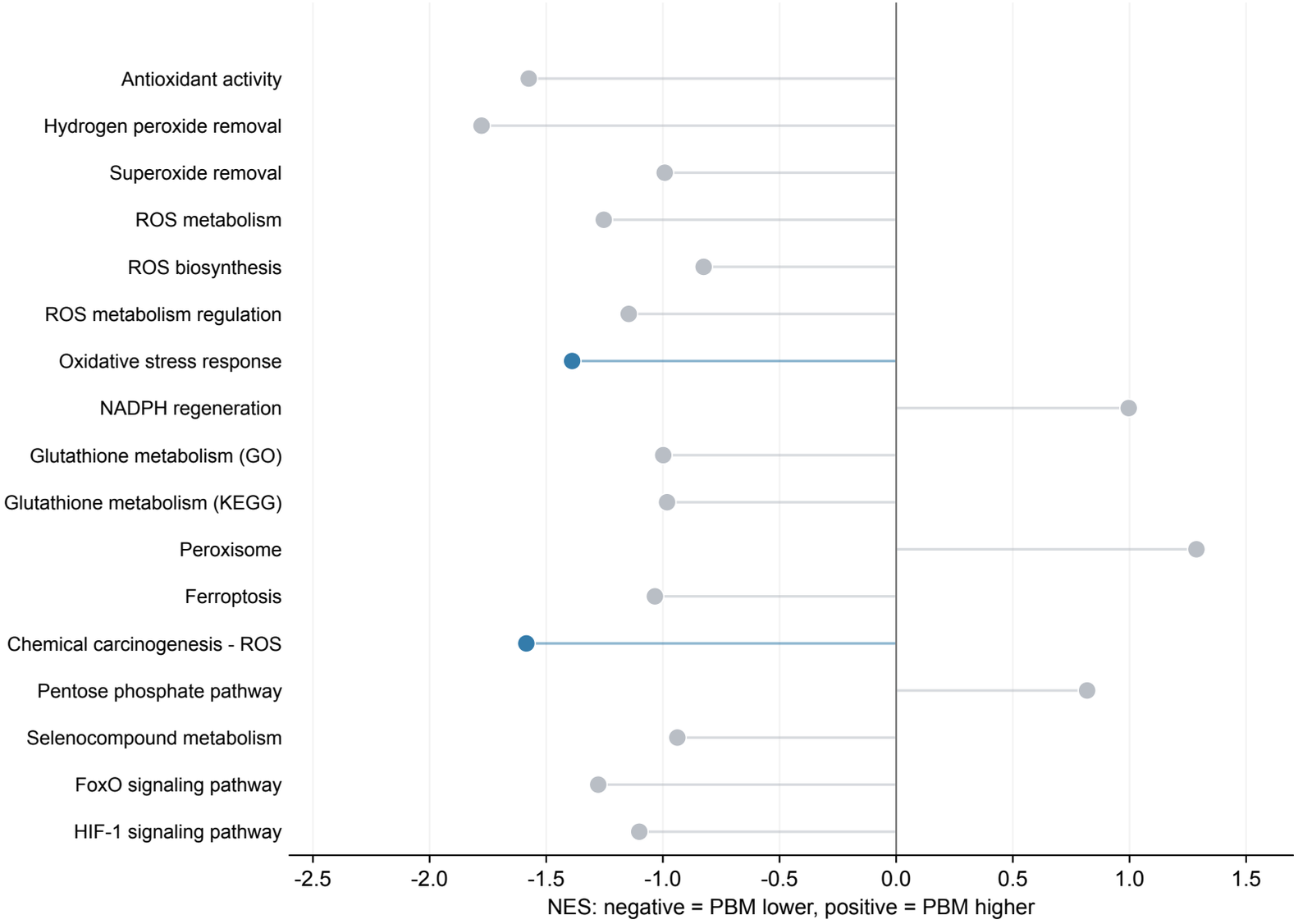
**GSE325653 | ROS / antioxidant pathway context**
**Prefrontal cortex, 1064 nm | deposited GEO DESeq2 statistics**




| | Gene-label q | Genes | P-rank | Sample q |
|---|---|---|---|---|
| Antioxidant activity | 0.079 | 59 | 43 | 0.551 |
| Hydrogen peroxide removal | 0.0641 | 11 | 39 | 0.512 |
| Superoxide removal | 0.711 | 14 | 229 | 0.573 |
| ROS metabolism | 0.323 | 88 | 110 | 0.725 |
| ROS biosynthesis | 0.877 | 19 | 279 | 0.983 |
| ROS metabolism regulation | 0.413 | 144 | 149 | 0.679 |
| Oxidative stress response | **0.0332** | 389 | 29 | 0.482 |
| NADPH regeneration | 0.701 | 20 | 226 | 0.85 |
| Glutathione metabolism (GO) | 0.701 | 62 | 223 | 0.988 |
| Glutathione metabolism (KEGG) | 0.716 | 61 | 236 | 0.968 |
| Peroxisome | 0.287 | 82 | 96 | 0.72 |
| Ferroptosis | 0.673 | 38 | 204 | 0.619 |
| Chemical carcinogenesis - ROS | **0.00968** | 204 | 22 | 0.624 |
| Pentose phosphate pathway | 0.895 | 29 | 291 | 0.619 |
| Selenocompound metabolism | 0.759 | 17 | 246 | 0.705 |
| FoxO signaling pathway | 0.264 | 123 | 89 | 0.72 |
| HIF-1 signaling pathway | 0.536 | 102 | 168 | 0.661 |

Coloured points: gene-label q < 0.05; grey: not significant. Both q columns are corrected over the full eligible set collection in this dataset.
P-rank is within all sets. KEGG context terms include broad redox signaling, not exclusively ROS scavenging. Sample-score and GSEA tests have different nulls.